

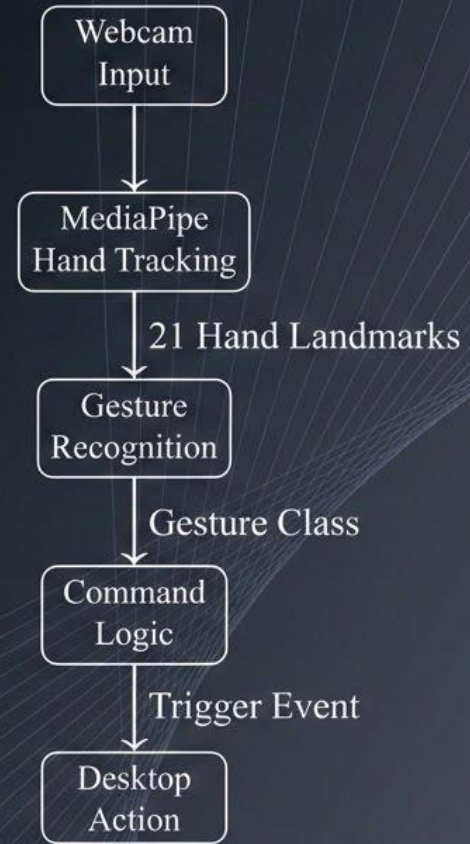
The background image is a close-up, high-contrast photograph of a blacksmith at work. A hammer is captured mid-swing, striking a piece of metal that is glowing with intense orange and yellow heat. Bright sparks are being ejected from the point of impact. The scene is set in a dark, industrial environment, likely a forge.

F.O.R.G.E.: Fine-tuned Optical Reconstruction for Gesture Extraction

Robert Teal, Jared Lillie, Seth McKinzie, Ryan Larrison

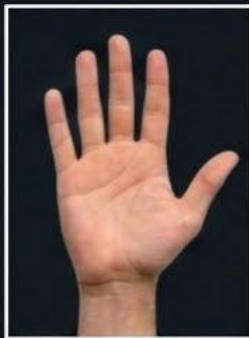
Project Overview

- A custom hand gesture interaction system
- Convert gestures into user commands
 - Toggle a desk lamp
 - Switch desktops
 - Open gemini in a browser



Data Collection

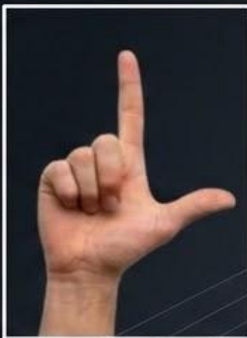
- Brainstormed meaningful gestures
- Collected our own data (~1000 images total)



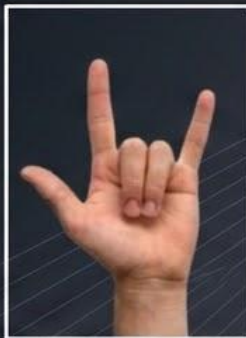
OPEN PALM



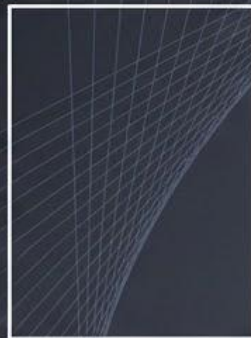
CLOSED FIST



L SHAPE



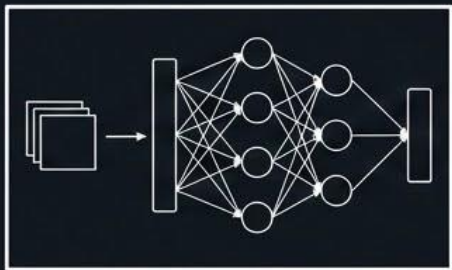
I LOVE YOU GESTURE



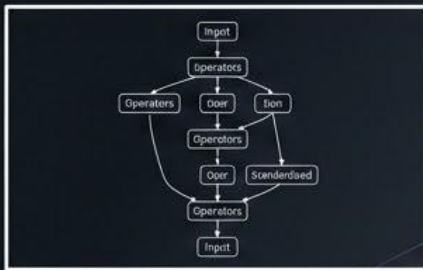
NO HAND

Training & Optimization

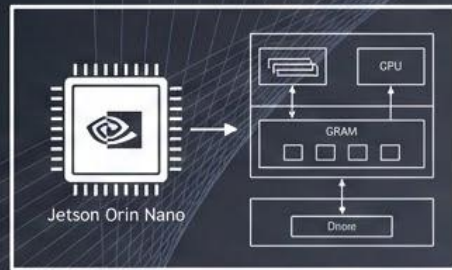
- 80/10/10 training/validation/test split for 10 epochs
- Increased model performance with TensorRT optimizations



Standard Network
(Mediapipe)

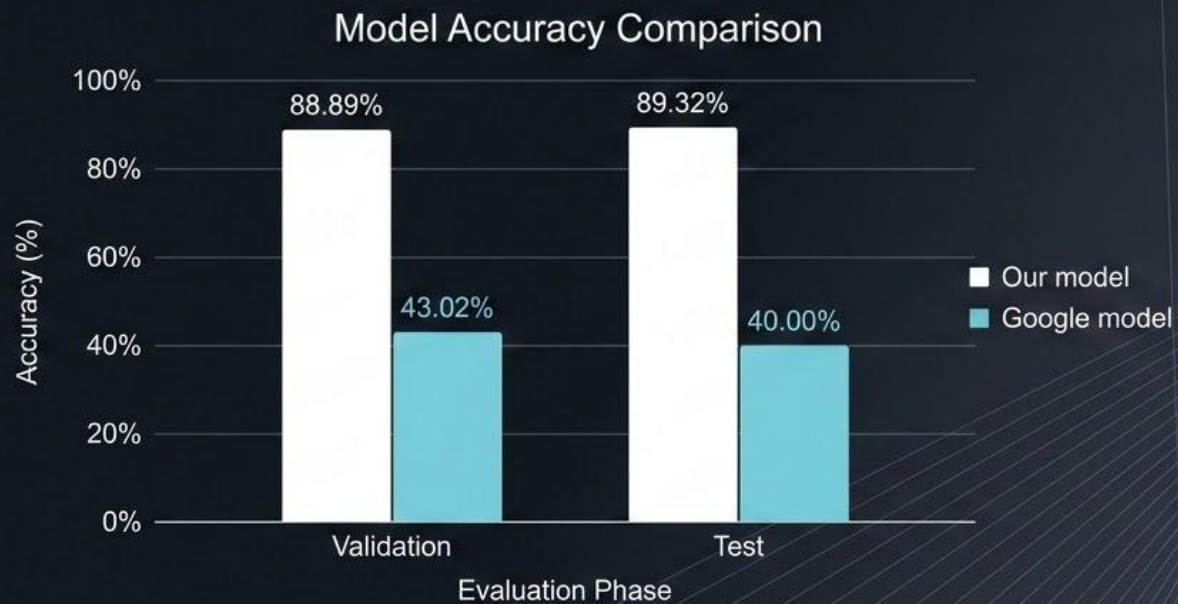


ONNX Format



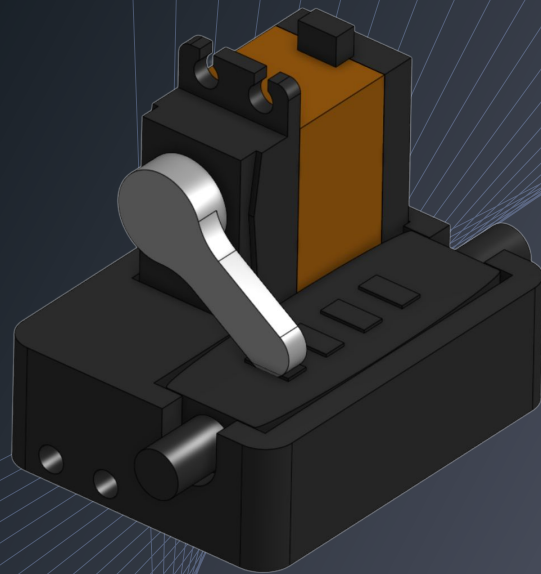
ENGINE Format
(Jetson Orin Nano)

Evaluation



Actions

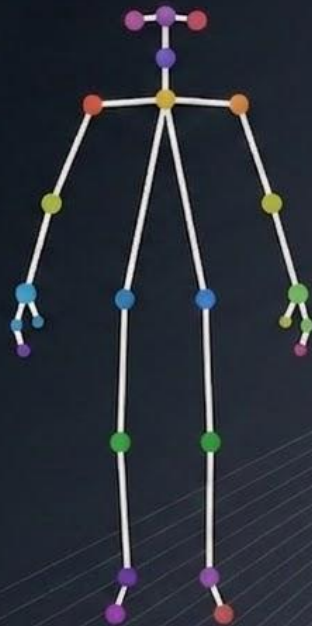
- Desk Lamp
 - Short pulse command sent over USB
 - Arduino Uno + 6V battery pack + torque servo + 3D printed mount
- Switch Desktops
 - Sends a system hotkey
- Open Gemini
 - Spawns Firefox with a custom URL



3D printed mount for
servo + lamp controller

Future Work

- Expand to full body tracking
- Implement LIDAR or similar light-independent gesture detection



Full body tracking



LIDAR gesture detection

The background features a series of thin, light blue lines that create a sense of depth and perspective. These lines originate from a vanishing point on the right side of the image and fan out towards the left, forming a grid-like pattern that recedes into the distance. The overall color palette is dark, with the lines providing a subtle contrast.

Demo